I love this direction. What you're describing is much more than a training-plan generator: it's a **personalized coaching platform for downhill mountain bike racers**, built around one flexible system that adapts to each athlete.

The key product principle should be:

One coaching engine. One shared training methodology. A unique plan for every rider.

You don't want to build a separate program for beginners, elite racers, riders without a gym, and athletes with only three days available. You want an app that understands those differences and adjusts the program automatically.

1. The app concept



DH Coach

Working name · Personalized performance coaching for gravity athletes

The app should help a rider answer five questions:

- What should I train today?
- Why am I doing this session?
- How should the workout adapt to my level and equipment?
- Am I getting fitter, stronger, and faster?
- What should change as my next race approaches?

The user experience should feel like having a personal coach, not filling out a spreadsheet.

2. The product architecture

I would build the app around four connected layers.

1. Athlete profile

Experience, goals, age, availability, equipment, injuries and preferences

↓

2. Coaching engine

Rules, training phases, exercise library, workload limits and progression

↓

3. Personalized plan

Daily sessions, gym workouts, bike skills, conditioning and recovery

↓

4. Feedback and adaptation

Workout completion, fatigue, progress and automatic plan adjustments

This separation matters. The coaching engine defines what makes a good training program. The personalization layer decides which version is appropriate for a particular athlete. The feedback layer adjusts it over time.

AI can help explain sessions, answer coaching questions, and suggest substitutions, while explicit rules govern progression, safety limits, and training load.

3. Design the onboarding system

This is the most important part of the app. Instead of asking every possible question at once, use a **progressive athlete assessment**: ask essential questions first, then request more detail when it improves the plan.



Step 1 — Who is the athlete?

- Age or age bracket
- Riding discipline: DH, enduro, freeride, or mixed
- Racing experience and competition category
- Main goal: fitness, speed, consistency, or first race

Purpose: establish appropriate training and safety boundaries.

Step 2 — What is their current level?

- Riding frequency and recent training history



- Gym experience and current strength-training routine
- Technical confidence and access to trails
- Current conditioning and recent performance

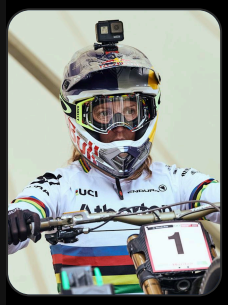
Purpose: establish the starting workload instead of assuming every athlete is intermediate.



Step 3 — What resources are available?

- Gym, home gym, or bodyweight only
- DH trails, bike park, pump track, or indoor trainer
- Training days and available time per session
- Coaching, training partners, and equipment constraints

Purpose: produce a plan the athlete can actually follow.



Step 4 — What are they preparing for?

- First race date or target event
- Number of races and priority events
- Current training phase
- Time available before competition

Purpose: work backward from the race calendar to create appropriate training blocks.



Step 5 — How is the athlete recovering?

- Sleep and perceived energy
- Training soreness and fatigue
- Previous injuries or current limitations
- Work, school, and other life demands

Purpose: determine whether to progress, maintain, modify, or recommend rest.

The onboarding rule

Don't force users to complete a 30-question form before seeing value.

Start with perhaps 8–10 essential questions. Generate a provisional plan, then ask optional questions that improve exercise selection, training intensity, and adaptation. Never require an athlete to disclose unnecessary medical information; collect only what is needed to support safe training.

4. Build a flexible athlete profile

Every answer should influence one or more specific parts of the program.

Profile variable	What the app changes

Age and development stage	Training suitability, supervision, and recovery
Experience level	Exercise complexity, volume, and progression
Days available	Weekly schedule and session distribution
Gym equipment	Exercise selection and alternatives
Riding access	Trail sessions versus alternative conditioning
Race date	Training phase and taper timing
Current fitness	Starting workload and conditioning
Recovery feedback	Daily intensity and workload adjustments
Goals	Relative emphasis on strength, power, skills, or endurance

A crucial design distinction: **training level is not the same as racing level**. An expert rider may be inexperienced in the gym. A beginner racer may have years of strength-training experience. The app should model these separately.

Similarly, a previous injury should not automatically trigger a diagnosis or rehabilitation program. It should inform exercise constraints and, where appropriate, prompt the athlete to seek professional guidance.

5. The coaching engine: how personalization actually works

Avoid relying on a single AI prompt to invent a complete training plan from scratch every time. That can lead to inconsistent workouts, unrealistic workloads, and unsafe progression.

Instead, use a structured exercise library and training rules.

Example: one goal, three personalized workouts

All three target lower-body strength, but the session changes according to the athlete's experience and resources.

Athlete A — Beginner Home

35–45 minutes

- Goblet squat or chair squat
- Hip bridge
- Supported split squat
- Dead bug

Focus: learn movement patterns and establish consistency.

Athlete B — Intermediate Full gym

50–65 minutes

- Squat variation
- Romanian deadlift
- Bulgarian split squat
- Calf raises and trunk stability

Focus: progressive strength development.

Athlete C — Advanced Full gym

55–70 minutes

- Individualized heavy compound lift
- Posterior-chain strength
- Unilateral strength
- Carefully dosed explosive work

Focus: maximize useful force while managing fatigue and preserving riding quality.

The app should choose a workout using a sequence like this:

1. Identify the athlete's training phase and target adaptation.
2. Filter exercises by equipment, experience, and stated limitations.
3. Select an appropriate starting volume and intensity.
4. Fit the session into the athlete's available time.
5. Check that the session is compatible with the rest of the week.
6. Present the workout with instructions, alternatives, and a clear purpose.
7. Use completion and recovery feedback to guide the next session.

Each exercise should have structured metadata: movement pattern, muscle groups, equipment, difficulty, coaching cues, contraindication flags, progression, regression, sets, reps, rest, and intended training effect.

This library becomes one of your app's most valuable assets.

6. The app should adapt every week

A genuinely personalized coach doesn't just generate a plan once. It learns from how the athlete responds.

For example:

Adaptive coaching example

- **The athlete completes all prescribed sets**

Technique is solid, recovery is good, and the session feels appropriate. The app may progress repetitions or load at the next planned opportunity.

The athlete feels unusually fatigued

The app reduces or replaces demanding work and checks whether the fatigue persists. It doesn't automatically compensate by adding more work later.

The athlete misses a training day

The app intelligently reschedules the priority sessions without cramming missed workouts into the remaining days.

Race day approaches

The app adjusts the training phase, reduces accumulated fatigue, and preserves short, familiar efforts that help the athlete feel sharp.

For the first version, use simple, transparent rules. Later, you can add wearable data, performance trends, automated load recommendations, and more sophisticated AI coaching.

7. What I would build for the MVP

Don't start by building every possible feature. Start with a product that can reliably generate and manage a personalized preseason plan.

Feature 1 — Athlete onboarding

Profile, training history, goals, equipment, schedule, and race dates.

Feature 2 — Personalized training calendar

Weekly schedules with gym, riding, conditioning, mobility, and recovery sessions.

Feature 3 — Workout details

Sets, reps, rest, exercise demonstrations, coaching cues, and equipment-based alternatives.

Feature 4 — Training log

Session completion, perceived effort, fatigue, soreness, and progress.

Feature 5 — AI coach

Explain the plan, answer training questions, suggest approved substitutions, and help athletes understand their progress.

8. A practical technical architecture

For a web-first MVP, I'd consider this setup:

Component	Suggested approach
User interface	Responsive web app; Next.js or a comparable framework
Database and authentication	Supabase with PostgreSQL
Training-plan engine	Structured rules and templates in application code
AI coaching	An LLM API with constrained, structured inputs and outputs
Exercise library	Database-backed exercise catalog with demonstrations
Analytics	Training history, adherence, workload, and progress metrics

Keep the athlete profile, plan version, workout prescriptions, completed sessions, and feedback as separate records. That makes it possible to regenerate future sessions without losing what was originally prescribed.

Personal data should be protected with appropriate access controls. AI-generated recommendations should pass through validation rules before being saved or shown to the athlete.

9. Let's define what we're actually building

There are two decisions that will shape the entire project:

- **Product scope:** are we building a DH-specific coaching product first, with room to expand into enduro and other disciplines, or a multi-discipline platform from day one?
- **Build approach:** are you looking for a working prototype you can test immediately, or the architecture and implementation plan for a production-ready app?

Define your first version

Which riders will it serve?

- ○ Downhill racers only
- ◉ DH first, expandable to enduro and other disciplines
- ○ DH, enduro, XC, and other disciplines from launch

What do you want to build first?

- ◉ A working interactive prototype I can test
- ○ A production app architecture and implementation plan
- ○ A no-code MVP I can launch quickly

Who is the intended customer?

- (•) Individual riders training independently
- ( ) Coaches managing multiple athletes
- ( ) Both individual riders and coaches

Where should people use it?

- (•) Web app, usable on desktop and mobile
- ( ) Native mobile app
- ( ) Both web and mobile apps

Define my app and next steps

My recommendation is to start with **DH first, individual riders, and a working web prototype**. Validate that the onboarding produces sensible plans for very different athletes before expanding the product. That will give you a solid coaching system to build on rather than a beautiful app that generates generic workouts.